Equal forces can have unequal effects

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Force pair / ES: par de fuerzas / PT: par de forças

A-on-B and B-on-A: equal and opposite.

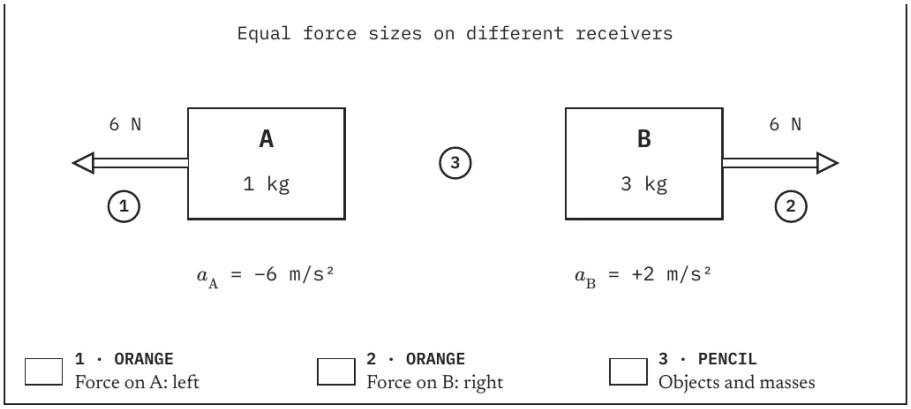
Also on your note page: Receivers, Mass effect, System.

B READ THE RELATIONSHIP

F(A on B) = −F(B on A); each receiver has its own a = F/m

As you read, find: on A · on B · same strength.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

Cart A has mass 1 kg ; cart B has mass 3 kg . Their interaction pushes A left and B right with 6 N each. Find both accelerations.

Given m A = 1 kg m B = 3 kg right is +

Rule F = ma on each cart. No other horizontal forces.

Work a A = −6/1 = −6 m/s² a B = +6/3 = +2 m/s²

Answer −6; +2 m/s² A has 3 times B's acceleration size.

E SAVE FOR THE FINAL MEETING

Earth and fictional XJ pull on each other equally at the same instant. Their very different masses give different accelerations; the pair does not cancel in XJ's own force diagram.

Read the physics: Force pair

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

If A pushes B, B pushes A at the same instant. The forces have the same strength. Swap the names to find the other receiver.

02 HOW IT WORKS

Draw one diagram for each receiver. Put only the force on that receiver in its diagram. These two arrows cannot cancel on one object.

03 WHAT TO CHECK

For the same force size, the smaller mass has the larger acceleration. Balanced forces on one object are a different idea.

04 TRY THE EXAMPLE FROM YOUR NOTES

Cart A has mass 1 kg ; cart B has mass 3 kg . Their interaction pushes A left and B right with 6 N each. Find both accelerations. −6; +2 m/s² A has 3 times B’s acceleration size.

Model check: F(A on B) = −F(B on A); each receiver has its own a = F/m.

05 CONNECT TO XJ

Earth and fictional XJ pull on each other equally at the same instant. Their very different masses give different accelerations; the pair does not cancel in XJ's own force diagram.

SAY IT BEFORE YOU WRITE

Do your two arrows end up on the same object? ___ pushes on ___. Its partner is ___ pushing on ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Force pair / ES: par de fuerzas / PT: par de forças

Say it in any language. Draw or use a few words. Use the notes to check your idea.

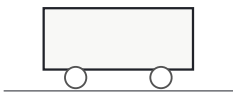
Meaning	How can you tell?
To me, force pair means …	Earth pulls on an apple. Where would you draw the partner force? I draw the partner force on ___, pointing toward ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

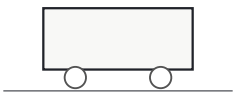
Label both receivers first. Draw equal force arrows on different objects, pointing toward one another for gravity.

Labels you can use: on A · on B · same strength.

Receiver A



Receiver B



Tell a partner: ___ pushes on ___. Its partner is ___ pushing on ___.

On assignment: Force pair

SECTION E / BACK / FINAL MEETING ONLY Name _____

Earth pulls on XJ. A public animation shows both gravitational arrows on XJ, with the caption “they cancel.”

PICK A JOB / DELIVER ONE PRODUCT

☐ **Simulation engineer**

Build two object panels. Put each gravitational arrow on its receiver so the animator can render the two motions.

☐ **Science graphics producer**

Prepare a split-screen caption: name who pulls whom and explain why equal force does not mean equal acceleration.

Word bank: on A · on B · same strength.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”

Editor’s check: Choose the combined system, then explain what is internal to that system.